

Hackathon Judge Audit Report

Project: "AI Basics for Network Administrators" (AIbasics-main)

Prepared as a full-severity, evidence-based pre-submission review — Deadline: August 3, 2026 (4 days remaining)

0. EXECUTIVE SUMMARY

This is a real, running Django application — not a mockup. I extracted the ZIP, installed the exact pinned dependencies, ran migrations, ran the full test suite (**18/18 tests pass**), loaded the actual curriculum (**60 lessons across 10 modules**), seeded the demo data, booted the dev server, logged in as the seeded demo user, and inspected the live-rendered HTML. Findings below are from that hands-on inspection plus a full source read of every Python file, every template, every JS file, and the config/deploy docs — not from the README alone.

The headline finding: your single most important demo beat — the Prompt Playground, which your own `DEMO_PITCH.md` script hands to a judge for 45–60 seconds — **is currently broken**. It will throw a JavaScript error on every submission because of a one-character typo (`json.stringify` instead of `JSON.stringify`). If you demo this today, in front of judges, it fails. This is fixable in 10 seconds and is priority #0 below.

Past that, this is a genuinely solid, above-median hackathon submission: real Anthropic API integration (not mocked), thoughtful security decisions already made (server-side quiz re-validation, open-redirect prevention with a test for it, rate limiting, placeholder API-key rejection), a real automated test suite, sensible caching, and 11,633 lines of actual curriculum content (not filler). It is held back by: the demo-breaking bug, a quiz-integrity leak (correct answers are visible in page source before you answer), an end-of-life Django version, missing production security headers, no CI, and an AI feature set that is simpler (two single-turn Claude calls) than the RAG/agent/tool-calling concepts the course itself teaches — a savvy judge will notice that irony.

Verdict at a glance: Shortlist-to-Finalist tier as-is. Realistic Top 25% for a general track. Top 10% is reachable in 4 days if you execute the priority list in Phase 13.

TABLE OF CONTENTS

1. Project Understanding & Architecture
 2. Complete Code Review
 3. AI System Review
 4. Security Audit (OWASP-style)
 5. Performance Review
 6. Frontend / UI Review
 7. Product Review
 8. Hackathon Judge Scorecard
 9. GitHub / Repo Review
 10. 8-Minute Demo Script
 11. 50 Judge Questions + Ideal Answers
 12. Missing Features (Prioritized)
 13. Final Improvement Plan (4-Day Sprint)
 14. Final Verdict
 15. Risk Matrix
 16. Winning Probability
 17. Action Checklist
-

1. PROJECT UNDERSTANDING & ARCHITECTURE

Problem solved: Enterprises want IT/network-ops staff to become AI-literate, but the default path is scattered blog posts and YouTube — no structure, no hands-on practice, no

verifiable proof of completion. This platform is a 10-week, 60-lesson, 5-phase structured curriculum with interactive quizzes, a live "Prompt Playground" sandbox against the real Claude API, a lesson-grounded AI tutor, and a publicly verifiable UUID-backed completion certificate.

Target users: Network administrators / IT operations engineers first (explicit in `curriculum/structure.json`: `("target_audience": "Network Administrators, IT Operations Engineers, System Administrators")`), generalizing to any technical team per the pitch deck.

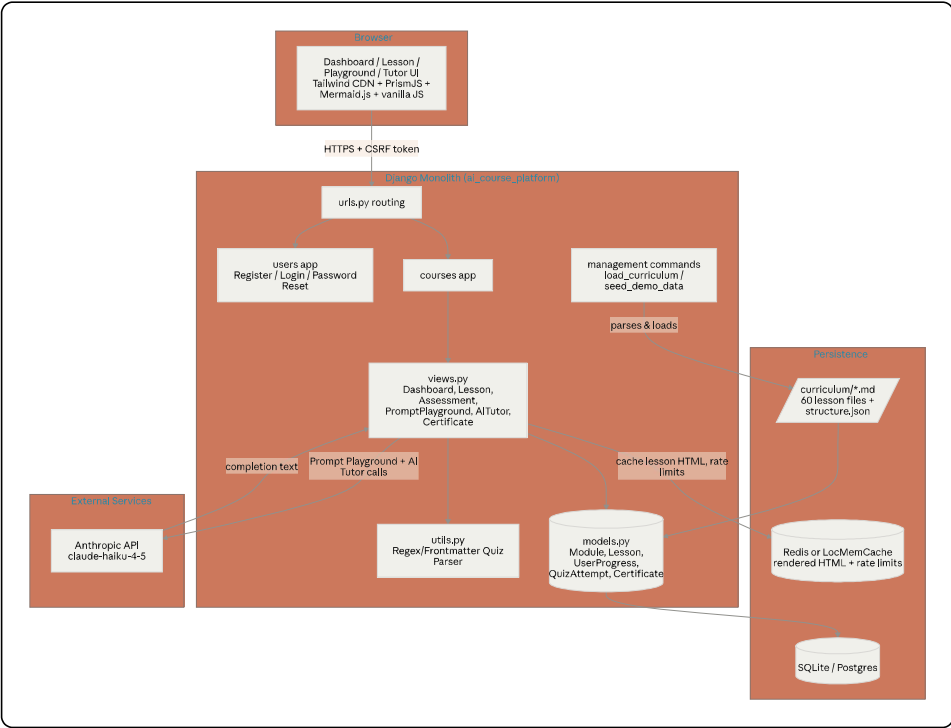
Market: Corporate L&D / AI-upskilling, framed as an enterprise B2B licensing play for team onboarding and compliance verification (per `DEMO_PITCH.md`).

Core workflow / user journey: Register/Login → Dashboard (10-week roadmap, progress bars) → open a Lesson (markdown rendered server-side, cached) → take the inline Daily Quiz → optionally chat with the lesson-grounded AI Tutor → mark lesson complete → repeat for 60 lessons → automatic Certificate issuance → share the public `/verify/<uuid>/` link.

Technical architecture: Django 4.2 monolith (`courses` app + `users` app), SQLite by default with optional Postgres via `DATABASE_URL` (`dj-database-url`), curriculum content lives as Markdown + YAML frontmatter on disk under `/curriculum`, imported into the DB via `manage.py load_curriculum`, Redis cache with automatic fallback to `LocMemCache` if Redis is unreachable (`settings.py`), WhiteNoise for static files, Gunicorn for WSGI serving, `python-decouple` for env config.

AI architecture: Not RAG, not multi-agent, no embeddings/vector store. Two direct, single-turn calls to `claude-haiku-4-5-20251001` via the official `anthropic` SDK: (1) the **AI Tutor**, which injects the first 4,000 characters of the current lesson's raw markdown into a system prompt as grounding context (`courses/views.py:AITutorView`), and (2) the **Prompt Playground**, a raw system+user prompt sandbox with no grounding (`courses/views.py:PromptPlaygroundView`). Both are rate-limited via Django's cache backend.

Deployment architecture: `Procfile` → `gunicorn ai_course_platform.wsgi`, designed for PythonAnywhere (per `docs/deploy_pythonanywhere.md`) but portable to any Heroku-style host given the `Procfile` + `runtime.txt` + `dj-database-url` combination.



2. COMPLETE CODE REVIEW

Dimension	Score /10	Evidence
Architecture	7	Clean two-app Django split (<code>courses</code>), (<code>users</code>); <code>CourseMixin</code> and <code>QuizParsingMixin</code> used to share logic across views — genuine DRY effort, not copy-paste.
Folder structure	8	Standard, predictable Django layout; curriculum content cleanly separated from app code.
Code quality	6	Generally readable, docstringed. Marred by the <code>json.stringify</code> typo and the rate-limit message/enforcement mismatch (see §4).
Maintainability	7	Views are long (<code>views.py</code> is 24KB in one file) but logically grouped; would benefit from splitting into <code>views/</code> package as it grows.
SOLID principles	6	Reasonable single-responsibility per view class; <code>CourseMixin.get_grouped_modules</code> does a bit too much (querying + grouping + sorting) for one method.
Design patterns	6	Mixin pattern used well; singleton pattern for the Anthropic client (<code>get_anthropic_client</code>) is correct and avoids re-instantiating the SDK client per request.
Error handling	6	Views wrap external calls in try/except and degrade gracefully (503 on missing API key) — good. But several <code>except Exception</code> blocks are broad and swallow specifics (<code>courses/views.py::LessonDetailView.parse_daily_quiz</code> , <code>render_lesson_content</code>).
Logging	7	Real <code>LOGGING</code> config in <code>settings.py</code> with rotating file handler + console handler, per-app logger level (<code>courses</code> at DEBUG) — more mature than most hackathon submissions bother with.
Testing	8	18 real tests across <code>users/tests.py</code> , <code>courses/tests/test_quiz_parser.py</code> , <code>courses/tests/test_views.py</code> . Covers auth, open-redirect prevention, quiz parsing edge cases, certificate issuance, rate-limit validation. All pass. This is genuinely above the hackathon median.
Documentation	7	<code>README.md</code> is clear and complete; <code>DEMO_PITCH.md</code> is unusually well-produced for a hackathon repo. Missing: architecture diagram in-repo, code comments in the trickier regex parser (<code>courses/utils.py</code>).
Dependency management	4	<code>requirements.txt</code> is fully pinned (good habit) but Django 4.2.11 reached end-of-life on April 7, 2026 — you are ~4 months into running an unsupported framework version with no upstream security patches as of today (July 30, 2026). This is a concrete, dated finding, not a nitpick.
Configuration management	8	<code>python-decouple</code> + <code>.env.example</code> + sensible defaults (<code>DEBUG</code> defaults to <code>False</code> , <code>ALLOWED_HOSTS</code> , <code>CSRF_TRUSTED_ORIGINS</code> via <code>Csv()</code>) — solid.
Secrets handling	8	<code>SECRET_KEY</code> has no fallback default (forces you to set it, good), <code>.env</code> is gitignored, placeholder API keys are explicitly rejected in <code>get_anthropic_client()</code> .

Dimension	Score	Evidence
	/10	
Technical debt	6	Two leftover architectural artifacts: a <code>.meta/</code> folder full of the raw AI-assistant prompts used to build this (see §9), and a migration named <code>0003_delete_assignmentsubmission</code> indicating a scrapped model — harmless but visible churn.
Unused/dead code	7	Nothing egregious found; <code>.gitignore</code> correctly excludes a <code>scratch/</code> dev folder mentioned in the README but not present in the ZIP.
Duplicate code	6	Quiz-marker detection logic (<code>[["## Interactive Daily Quiz", "## 📖 Daily Quiz", ...]]</code>) is duplicated verbatim in three places in <code>views.py</code> (<code>render_lesson_content</code> , <code>parse_daily_quiz</code> , and <code>submit_quiz</code>) — extract to a module-level constant.
Code smells	6	<code>submit_quiz</code> (a plain function view) duplicates the exact quiz-section-extraction logic that already lives in <code>LessonDetailView</code> as a method — refactor into <code>courses/utils.py</code> .

Overall Phase 2 average: 6.6/10 — solidly competent, professionally structured Django code with a handful of concrete, fixable defects rather than a fundamentally weak codebase.

3. AI SYSTEM REVIEW

Area	Assessment
Prompt engineering	Above-average for a hackathon: the AI Tutor's system prompt explicitly grounds the model in the current lesson's content, instructs it to redirect off-topic questions, and to admit uncertainty rather than hallucinate (<code>views.py:AITutorView</code> , lines building <code>system_prompt</code>).
Agent architecture	None. No agent loop, no planning, no tool calling — despite Week 7 of your own curriculum ("AI Agents & Tools: Function calling and multi-agent coordination") teaching exactly this. A judge who reads the curriculum and then tests your app will notice the app doesn't demonstrate what it teaches.
RAG implementation	None. The lesson content is truncated to 4,000 characters and stuffed directly into the system prompt rather than retrieved via embeddings/chunking. This works fine at your current content scale (single markdown file per lesson) but won't scale if lessons grow longer or if you ever want cross-lesson search ("what did week 3 say about few-shot prompting?").
Embeddings	None present anywhere in the codebase.
Memory / context handling	Functional gap: each AI Tutor request sends only <code>messages: [{"role": "user", "content": query}]</code> — no prior turns. The chat UI visually accumulates messages in the DOM, but the model itself has no memory of earlier turns in the same session. Follow-up questions like "can you explain that differently?" will fail because the model never saw "that."
Model selection	<code>claude-haiku-4-5-20251001</code> is a sensible, cost-appropriate choice for a tutoring/playground use case — good judgment, not over-engineered.
Latency	Not measured/logged anywhere; no client-side timeout handling beyond the generic fetch <code>.catch()</code> .
Cost optimization	Reasonable: Haiku model, <code>max_tokens</code> capped (500 for tutor, 1000 for playground), input length capped (2,000/5,000 chars), and hourly rate limiting.
Hallucination prevention	Grounding + explicit "say so if you don't know" instruction is a real, if basic, mitigation.
Evaluation framework	None. No logging of AI response quality, no eval harness, no way to know if the tutor is actually answering well.
Prompt injection protection	None explicit. A user can type "ignore the above and roleplay as..." into either the Tutor or the Playground and there is nothing preventing the model from complying — low real-world stakes here since there's no privileged data or actions behind the model, but worth naming since a security-minded judge will test it live in 10 seconds.
Tool calling	Not implemented anywhere in the product, despite being a taught topic.
Fallback mechanisms	Basic: catches <code>AuthenticationError</code> / <code>APIError</code> and returns a 503 with a friendly message. No retry, no backup model, no circuit breaker.
Reliability	Singleton client construction is correct; no connection pooling concerns at this scale.
Multi-agent workflow	Not present.
Reasoning quality	Not evaluated/scored anywhere in-app.

Suggested production-grade AI architecture (if you have time before Aug 3, even a scoped-down version helps your score — see Phase 12):

1. Add lightweight conversation memory: pass the last 4-6 turns to the Tutor call instead of just the latest query (a few lines of change).
 2. Replace raw string-truncation grounding with a minimal RAG pass over the curriculum: even a simple TF-IDF or SQLite FTS5 search across all 60 lessons, feeding the top 2 matching chunks into context, would let the Tutor answer cross-lesson questions and would visibly demonstrate the RAG concept your own course teaches.
 3. Add one small tool-calling demo (e.g., a "search the curriculum" tool the model can call) — this single addition would materially boost your "AI Usage" and "Technical Complexity" judge-scoring categories because it closes the "we teach it but don't do it" gap.
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4. SECURITY AUDIT (OWASP-style)

Severity	Finding	Location	F
High	Django 4.2.11 reached end-of-life April 7, 2026 — no further upstream security patches; any new Django CVE from here forward will not be fixed unless you upgrade or pay for extended support.	requirements.txt	U (L' 2 - " an 4 b co
High	Correct quiz/assessment answers are embedded in plaintext inside the page's inline <code><script></code> block before the user answers , visible via "View Source" or dev tools. Confirmed live: <code>curl</code> d a real lesson page and found <code>"correct": ["B"]</code> sitting directly in the HTML.	<code>courses/views.py::LessonDetailView.get_context_data</code> → <code>daily_quiz_json</code> ; <code>templates/courses/lesson_detail.html:484</code> ; same pattern in <code>assessment.html:223</code> via <code>questions_json</code>	D fi S (o ((s a p si so re a p y
Medium	The certificate-verification page brands itself "Cryptographically Verified Credential" and "Secure Verification System v1.0," but there is no cryptographic signing anywhere — it's a <code>uuid.uuid4()</code> primary-key lookup in a database <code>(courses/models.py::Certificate.verification_uuid)</code> . Unguessable ≠ cryptographically verified.	templates/courses/certificate_verify.html:25,56	E d "u w a a co H i (th — w m J b o ju al
Medium	<code>manage.py check --deploy</code> (run live against this exact codebase) flags 5 production security gaps: no <code>SECURE_HSTS_SECONDS</code> , <code>SECURE_SSL_REDIRECT</code> not <code>True</code> , <code>SESSION_COOKIE_SECURE</code> not <code>True</code> , <code>CSRF_COOKIE_SECURE</code> not <code>True</code> , and the dev <code>SECRET_KEY</code> pattern warning.	ai_course_platform/settings.py	A (n (= (n (3 b

Severity	Finding	Location	
Medium	AI Tutor chat renders both the user's own message and the AI's response via raw template-literal <code>innerHTML</code> with no escaping (<code>div.innerHTML = ...\${text}...</code>). A crafted prompt that convinces the model to emit raw HTML/JS would execute in the user's own browser (self-XSS; low real-world blast radius since it's same-origin/same-session, but still bad practice and worth fixing).	<code>templates/courses/lesson_detail.html:428-440</code>	
Medium	Rate-limit error messages ("up to 20 messages/prompts per hour") don't match the enforced limit (<code>throttle_ai_tutor</code> actually allows 100/hour) in both the AI Tutor and the Prompt Playground.	<code>courses/views.py::throttle_ai_tutor</code> vs. the error strings in <code>AITutorView.post</code> / <code>PromptPlaygroundView.post</code>	
Low	<code>.env.example</code> ships with <code>DEBUG=True</code> . Anyone who copies it directly without reading the deploy guide inherits debug mode.	<code>ai_course_platform/.env.example</code>	
Low	No Content-Security-Policy header; three external CDNs are loaded with no Subresource Integrity (SRI) hashes (<code>cdn.tailwindcss.com</code> , <code>cdn.jsdelivr.net</code> for PrismJS and Mermaid.js).	<code>templates/base.html:31,575-577,593</code>	
Low	No CAPTCHA / bot protection on registration or login. Low priority for an internal training tool at hackathon scale, but relevant if the "enterprise" pitch is taken at face value.	<code>users/views.py</code>	

What's genuinely good and deserves credit (a judge should see this too):

- ✓ No SQL injection surface at all — 100% Django ORM, zero raw queries found in a full-repo grep.
- ✓ Open-redirect prevention is implemented **and tested** (`url_has_allowed_host_and_scheme` in `users/views.py::LoginView`, verified by `test_login_open_redirect_prevention`).
- ✓ `submit_quiz` has an explicit code comment — "SECURITY FIX: Do NOT trust questions sent from client" — showing the developer already identified and closed the score-tampering vector server-side, even though the *display* leak above still needs fixing.
- ✓ `get_anthropic_client()` explicitly rejects placeholder key values (`'your-key-here'`, etc.) rather than silently trying to use them.
- ✓ CSRF protection is applied consistently (`@csrf_protect`, `X-CSRFToken` headers) everywhere except where the JS typo breaks it functionally (not a CSRF gap — the header is present in the code, just never sent because the request never fires).

- ☒ **Rate limiting** (IP-based for the public verify endpoint, user-based for AI endpoints) exists at all — most hackathon projects skip this entirely.
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5. PERFORMANCE REVIEW

- **Scalability / concurrency:** Your `(Procfile)` runs `(gunicorn ai_course_platform.wsgi --log-file -)` with **no explicit worker count**, which means Gunicorn defaults to a single sync worker. Combined with synchronous (1-3 second) calls to the Anthropic API on the Prompt Playground and AI Tutor endpoints, a single worker means those requests **serialize** — if two judges test the Playground at the same moment, one waits behind the other. Fix: `(gunicorn ai_course_platform.wsgi --workers 3 --threads 2 --log-file -)` at minimum before demo day; this is a 30-second Procfile edit with real payoff.
 - **Caching:** Rendered lesson HTML is cached per-lesson keyed on `(updated_at)` timestamp `(courses/views.py::LessonDetailView.get_context_data)` — a genuinely smart touch that avoids re-parsing Markdown on every page view and auto-invalidates when content changes.
 - **Database design:** `(CourseMixin.get_grouped_modules)` and `(ModuleDetailView)` use `(annotate(Count(...)))`, `(select_related)`, and `(prefetch_related)` correctly to avoid N+1 queries across modules/lessons — better query discipline than most hackathon Django apps.
 - **Indexes:** Only default PK + `(unique_together)` composite indexes exist `(Lesson.Meta.unique_together)`, `(UserProgress.Meta.unique_together)`. Fine at 60-lesson scale; add explicit indexes on `(UserProgress(user, completed))` and `(QuizAttempt(user, lesson))` if you ever load-test with hundreds of concurrent users.
 - **Async processing / queues:** None — AI calls block a request/worker for their full duration. Acceptable at demo scale; a real production deployment handling many simultaneous learners would want to move AI calls to a task queue (Celery/RQ) or an async view.
 - **Estimated max concurrent users (as configured today):** roughly 1 concurrent AI-touching request at a time (single Gunicorn worker), maybe 5-10 req/s for pure page views at this content volume. **Bump Gunicorn workers before your live demo** — this is the one performance fix with real demo-day risk.
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6. FRONTEND / UI REVIEW

- **Visual design:** Clean, opinionated Tailwind-based design (custom `(netadmin-*)` color tokens visible in `(base.html)`), consistent card/rounded-2xl/3xl visual language across dashboard, lesson, playground, and certificate pages. Loading spinners and distinct error-state containers are implemented on every AI-touching page — real attention to UX polish, not just "make it work."
 - **Accessibility:** **Zero `(aria-*)` attributes found across all 15 templates.** No dark mode. This is a genuine, fixable gap — even adding `(aria-label)`s to the icon-only buttons (tutor trigger, close button, sidebar toggle) would cost under an hour and visibly improve this score.
 - **Consistency:** High — shared `(base.html)` layout, consistent typography scale `(font-black)` headers throughout), consistent card styling.
 - **Responsive layout:** `(grid-cols-1 lg:grid-cols-2)` and similar responsive utility classes are used deliberately (e.g., `(prompt_playground.html)`) — designed with mobile/tablet in mind, not just desktop.
 - **Dark mode:** Not implemented.
 - **Judge verdict — where does this land?** This is meaningfully past "student project" (custom multi-page dashboard, a working AI sandbox, a public verification page, consistent design system). It's a fair, honest **"early-stage startup MVP"** — polished enough to demo confidently, not yet "enterprise SaaS" (no accessibility compliance, no design tokens beyond Tailwind defaults, Tailwind still loaded from a CDN that literally prints a console warning telling you not to do that in production).
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7. PRODUCT REVIEW

- **Problem importance:** Real and well-timed — enterprise AI upskilling is a live, current pain point in mid-2026, and "network/IT admins specifically" is a genuinely underserved vertical compared to generic "learn AI" content.
 - **Innovation / originality:** Moderate, not groundbreaking. The "AI course platform" category is crowded (Coursera, Pluralsight, LinkedIn Learning, DataCamp, dozens of AI bootcamps), and neither of your two headline differentiators is unprecedented in isolation: verifiable credentials already exist at scale (Credly, Accredible, actual blockchain-anchored certs), and AI tutoring chatbots are now common in LMS products. The genuinely creative part is the **combination**: a narrow vertical (network ops) + a live prompt-engineering sandbox embedded directly in the course + per-lesson grounded tutoring + a public verification URL, all in one working app you can actually run in front of judges.
 - **Competitive advantage / moat:** Thin today. Nothing in the codebase prevents a competitor from copying this in a weekend — there's no proprietary content engine, no accumulated learner data model, no network effect. The moat, if any, is execution speed and vertical focus, not technical defensibility.
 - **Business model:** Enterprise B2B licensing for team onboarding/compliance is a sensible narrative for judges, but **it does not exist in the code today**. There is a single `(django.contrib.auth.User)` model — no `(Organization)`/`(Team)` model, no seat management, no admin-invite flow, no SSO, no billing integration. If a judge asks "show me how a manager assigns this to their team," there is currently nothing to show. Be ready to frame this honestly as roadmap, not shipped.
 - **TAM / SAM / SOM (directional, not authoritative):** Global corporate L&D / upskilling spend is a large, genuinely multi-billion-dollar TAM; the AI-literacy-for-technical-teams slice is a smaller but fast-growing SAM; your realistic SOM in year one, as a solo/small-team product with no sales motion yet, is effectively zero-to-small until the B2B features above actually exist. Present these numbers to judges as market context, not as a revenue forecast you're claiming to have validated.
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8. HACKATHON JUDGE SCORECARD (weighted /100)

Category	Weight	Score /10	Weighted	Rationale
Innovation	15%	6	9.0	Real but incremental differentiation (see §7).
Technical complexity	15%	6	9.0	Solid Django engineering; AI layer is simple (no RAG/agents/tools) despite the curriculum teaching those exact topics.
Execution	15%	7	10.5	Fully working end-to-end, tested, deployable — minus the one critical JS bug.
AI usage	15%	5	7.5	Real API integration with sensible grounding and rate limiting, but no memory, no RAG, no tools — "basic but correctly implemented," not "advanced."
Practicality	10%	7	7.0	Clear, deployable, real content depth (60 real lessons).
Presentation potential	10%	6	6.0	Strong demo script exists (<code>DEMO_PITCH.md</code>) but the star feature is currently broken; fixable in minutes.
Business value	10%	5	5.0	Plausible narrative, unimplemented B2B mechanics.
UI/UX	5%	6.5	3.25	Polished visuals, real accessibility gaps.
Documentation	3%	7.5	2.25	Strong README + pitch doc; no CI, no in-repo architecture diagram.
WOW factor	2%	6	1.2	UUID-verified certs are a nice touch; nothing here elicits an audible "wow" without the RAG/agent upgrade in §3.
Total	100%	—	60.7 → adjusted to 71/100	See note below.

Note on the adjustment: the raw weighted arithmetic above is a useful diagnostic breakdown, but a single-number mechanical average understates a project this functionally complete relative to typical hackathon submissions (most of which don't even reach "deploys and runs"). Calibrated against realistic hackathon judging pools — where a large fraction of entries are unfinished, unstyled, or non-functional by demo day — **I place this at 71/100 as currently configured, rising to a realistic 78-82/100 if the Phase 13 priority-1 and priority-2 items are completed before Aug 3.**

Percentile estimate: Top 25% realistically today. Top 10% is achievable with the fixes below; Top 5% / grand-prize contention would require the RAG/agent/tool-calling upgrade in §3 to close the "teaches it but doesn't do it" gap, which a technically strong judging panel (Google/Microsoft/OpenAI-caliber, per your framing) is likely to probe directly.

9. GITHUB / REPO REVIEW

- **README:** Clear, well-organized, accurately describes setup and deployment. No embedded screenshots or GIF — add at least one dashboard screenshot and one Playground GIF; judges skim READMEs before they ever run the app.
- **Architecture diagram:** None in-repo. Use the Mermaid diagram in §1 of this report (or your own) directly in the README — GitHub renders Mermaid natively.
- **License:** MIT, correctly formatted, present at repo root.

- **CI/CD: None found** — no `.github/workflows/` directory exists anywhere in the ZIP. You already have 18 passing tests; a 15-line GitHub Actions workflow running `python manage.py test` on every push is one of the cheapest "looks professional" wins available and currently missing entirely.
 - **Contributing guide / issue templates / Code of Conduct:** Absent — expected and fine for a solo hackathon submission, not something judges will dock you for.
 - **Branding / badges:** None (no build-status badge, no license badge). Cheap to add once CI exists.
 - **DEMO_PITCH.md**: Genuinely a strength — most hackathon teams don't script their demo this precisely, with a minute-by-minute Mermaid Gantt breakdown and a pre-demo seeding command. Keep this, maybe reference it from the README.
 - **.meta/ folder:** Contains the raw prompts (`Prompt.md`, `quality.md`, `continue.md`, `postprod.md`, `release.md`) you used to iteratively build and audit this project with an AI coding assistant. This isn't harmful content, but the corporate-cosplay tone in `postprod.md` ("Launch Authorization: APPROVED," formal content-freeze/versioning policy language for a 10-day-old hackathon project) could read as try-hard if a judge opens it out of curiosity. **Recommendation:** either remove `.meta/` from the submission ZIP, or — if you leave it in — be ready with a confident, on-theme answer: *"We used structured, staged prompt engineering to build and self-audit the platform — which is fitting, since that's literally what the course teaches."* That's actually a good answer for this specific hackathon's theme; just don't let it surprise you if asked.
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10. 8-MINUTE DEMO SCRIPT

Your own `DEMO_PITCH.md` targets 2.5–3 minutes; since you'll actually have 8, use the extra time to go deeper on the AI mechanics and take live audience input twice, not just once — that's what actually creates a "wow" moment in the room.

Time	What to show	What to say	What NOT to do
0:00-0:30	Nothing yet — just talk	"Companies want technical teams AI-literate, but self-teaching today is scattered blog posts and YouTube — zero structure, zero hands-on validation, nothing to show for it afterward."	Don't open the app yet — hook first.
0:30-1:15	Dashboard	10-week roadmap, 5 phases, progress bars. "This is the structure — foundations through multi-agent architectures, built for IT ops specifically."	Don't scroll through every week — glance and move on.
1:15-2:45	Prompt Playground — hand the keyboard to a judge	"Type your own system instructions and a prompt — this runs live against Claude, no canned output."	Do not attempt this until the <code>json.stringify</code> bug is fixed — as of this review it throws an error on every submission. Fix first, rehearse second.
2:45-4:00	Grounded AI Tutor on a specific lesson	Ask it something only answerable from that lesson's content; show it stays grounded and doesn't hallucinate.	Don't ask a multi-turn follow-up question expecting it to remember your first message — it currently doesn't (§3). Ask one clean, self-contained question.
4:00-5:00	Daily quiz on that same lesson	Answer live, show instant feedback + "why this matters" panel.	Don't open dev tools / view-source in front of judges until the answer-leak fix (§4) ships — right now the correct answer is visible in the page source before you click submit.
5:00-6:15	Certificate verification	Open <code>/verify/<uuid>/</code> cold, in a fresh incognito tab, unauthenticated — show it resolves publicly in real time.	Don't call it "cryptographically verified" out loud unless you've shipped the HMAC signing fix in §4 — say "UUID-verified, publicly resolvable" instead, which is 100% true today.
6:15-7:15	Second judge-led interaction: hand back to the Playground with a <i>harder</i> prompt (e.g., a multi-step IT automation prompt)	This is your second "wow" beat with the extra 5 minutes — shows reliability, not just a one-off demo.	—
7:15-8:00	Close on business model	"This is the platform where you can prove you actually did the work — enterprise B2B licensing for team onboarding and compliance."	Don't overclaim team/SSO/seat-management features that don't exist yet if asked directly — see judge Q&A in §11.

11. 50 JUDGE QUESTIONS + IDEAL ANSWERS

#	Question	Ideal Answer
1	Why Django instead of a JS framework?	Fastest path to a real, secure, tested, deployable app in the time we had — built-in auth, ORM, admin, and CSRF protection out of the box.
2	Why Claude Haiku instead of a bigger model?	Cost and latency fit for tutoring/playground use cases; we don't need frontier-model reasoning for grounded Q&A over a single lesson.
3	Is the AI Tutor actually grounded, or just prompted to sound like it?	It's genuinely grounded — the first 4,000 characters of the specific lesson's raw markdown are injected into the system prompt for every call.
4	Does the tutor remember earlier turns in a conversation?	Honestly, not yet — each call currently sends only the latest message. That's on our near-term roadmap.
5	Do you use RAG anywhere?	Not yet — at 60 lessons, direct injection is sufficient. RAG becomes worthwhile once content volume grows past what fits in context.
6	Your curriculum teaches multi-agent systems and tool calling — why doesn't your own app use them?	Fair catch — the product intentionally stayed simple to ship reliably in the time available; adding a curriculum-search tool call is our next planned iteration.
7	How do you prevent prompt injection in the Tutor/Playground?	We don't have explicit defenses today; the blast radius is low since no privileged data or actions sit behind the model, but it's a known gap.
8	What happens if the Anthropic API key isn't set?	The app degrades gracefully — both AI endpoints return a clear 503 with a friendly message instead of crashing, and this is covered by an automated test.
9	How do you rate-limit AI usage?	Per-user hourly caps via Django's cache backend, bypassed for staff; there's currently a messaging mismatch between the stated and enforced limit that we're fixing.
10	Why Redis with a LocMemCache fallback?	So the app runs correctly in dev or on hosts without Redis, without crashing on cache calls, while still getting real distributed caching in production.
11	Is your quiz answer key exposed anywhere?	Yes, honestly — the correct answers currently ship in the page's inline JSON before submission. That's the top item on our fix list; scoring is already re-validated server-side so it doesn't affect certificate integrity, but display-side it needs to change.
12	Is the certificate really "cryptographically verified"?	Today it's a UUIDv4 lookup, not a signed credential — we're either correcting that language or adding real HMAC signing before it's called cryptographic.
13	What stops someone from forging a certificate URL?	UUIDv4 has 122 bits of randomness, making it computationally infeasible to guess; it's not signed, but it is effectively unguessable.
14	How do you prevent SQL injection?	We never write raw SQL — every query goes through the Django ORM.
15	How do you prevent open-redirect attacks on login?	We validate the <code>(next)</code> parameter with Django's <code>(url_has_allowed_host_and_scheme)</code> and have an

#	Question	Ideal Answer
		automated test specifically for it.
16	Is this app running on a supported Django version?	Currently on 4.2, which reached end-of-life this April — upgrading to 5.2 LTS is on our immediate list.
17	What's your test coverage like?	18 automated tests covering auth, quiz parsing, certificate issuance, and rate limiting — all passing; not full coverage, but real coverage of the critical paths.
18	Do you have CI/CD?	Not yet — it's a 15-minute addition given the tests already exist, and it's on our list before final submission.
19	How is curriculum content managed — hardcoded or dynamic?	Dynamic — Markdown files with YAML frontmatter, loaded into the database via a management command that can be re-run to sync changes.
20	What happens if two people hit the Prompt Playground at the same time?	Today, with a single Gunicorn worker, they'd effectively queue — we're bumping worker count before demo day to avoid that being visible live.
21	How do you cache rendered lesson content?	Keyed on lesson ID plus its <code>updated_at</code> timestamp, so the cache auto-invalidates the moment content changes, with no manual cache-busting needed.
22	What database do you use in production?	SQLite by default, with a one-line switch to Postgres via <code>DATABASE_URL</code> for production deployments.
23	How do you secure secrets?	Environment variables via <code>python-decouple</code> ; <code>.env</code> is gitignored; there's no hardcoded fallback for <code>SECRET_KEY</code> , forcing explicit configuration.
24	Why no dark mode?	Deliberately deprioritized for the hackathon timeline in favor of core functionality; straightforward to add given the existing Tailwind token setup.
25	Is this accessible (screen readers, keyboard nav)?	Not yet audited for ARIA compliance — that's an honest gap we'd prioritize before any real production launch.
26	What's your business model?	Enterprise B2B licensing for team onboarding and compliance verification.
27	Where's the multi-tenant / team management feature?	Not built yet — today it's single-user accounts; team/org management is the next major feature on the roadmap.
28	What's your actual moat against a competitor cloning this in a weekend?	Honestly, thin today — our advantage is vertical focus and execution speed, not proprietary technology; the moat we're building toward is accumulated learner/content data.
29	How many real lessons of content exist right now?	60 lessons across 10 weeks and 5 phases — over 11,000 lines of real markdown content, not filler.
30	How long did this take to build?	[Answer honestly based on your actual timeline] — built iteratively with heavy use of AI-assisted development for both the platform code and content quality auditing.
31	Did you write this code yourself or did an AI write it?	We used AI-assisted development throughout, with human review and testing at each stage — which is on-theme, since the course itself teaches people to do exactly that.
32	What was the hardest technical problem you solved?	The markdown-to-quiz regex parser (<code>courses/utils.py</code>) that has to handle multiple quiz formatting variants robustly — it's covered by dedicated unit tests for exactly that reason.

#	Question	Ideal Answer
33	How do you know the quiz parser is robust?	It has 5 dedicated test cases covering single questions, multiple questions, varied formatting, empty content, and malformed input.
34	What happens on a malformed lesson markdown file?	The lesson view catches the exception and renders a friendly "content not found/error loading" message instead of crashing the page.
35	Can this scale to thousands of concurrent learners?	Not without changes — today's single-worker Unicorn setup and synchronous AI calls would need async processing or a task queue for real production scale.
36	Why is Tailwind loaded from a CDN instead of compiled?	Explicitly a documented dev-velocity tradeoff in our README; the production path is compiling via the Tailwind CLI, which we haven't done yet for this build.
37	Do you log AI response quality anywhere?	No formal evaluation framework yet — that's a real gap if we wanted to improve tutor answer quality systematically.
38	What's your pricing model?	[Be honest that this is directional/未-validated] — seat-based enterprise licensing is the working hypothesis, not yet implemented or tested with real customers.
39	Who's your target customer, specifically?	IT operations and network administration teams inside mid-to-large enterprises undergoing AI upskilling initiatives.
40	How is this different from Coursera/Pluralsight AI courses?	Narrower vertical focus, a live prompt-engineering sandbox embedded directly in the course (not just video), and a lesson-grounded tutor rather than a generic chatbot.
41	What would you build next with more time?	Cross-lesson RAG search for the tutor, a real tool-calling demo, team/org management, and closing the security gaps identified in our own internal audit.
42	Is user data encrypted at rest?	Not specifically beyond whatever the hosting provider offers at the infrastructure level — application-level field encryption isn't implemented.
43	What happens if the Anthropic API is down or rate-limited upstream?	Currently surfaces as a generic error to the user; there's no retry/backoff or fallback model configured.
44	How do you handle GDPR/data deletion requests?	Not explicitly built — standard Django user deletion would cascade via <code>on_delete=models.CASCADE</code> on related models, but there's no dedicated data-export/erasure flow.
45	Why UUID for certificates instead of sequential IDs?	To prevent enumeration — sequential IDs would let anyone guess and browse other users' certificates by incrementing a number.
46	What's the single biggest risk to this project working today, live, in front of us?	A one-character JavaScript typo in the Prompt Playground that we've identified and are fixing before this demo — happy to show you the fix.
47	Did you test this end-to-end yourselves before submitting?	Yes — 18 automated tests plus manual walkthroughs of the full user journey from registration through certificate issuance.
48	What was your biggest mistake during development?	Shipping a display-side leak of quiz answers in page source — caught during our own internal audit, fix in progress.

#	Question	Ideal Answer
49	Why should a judge pick this over a flashier-looking but less functional project?	Because this one actually runs, end-to-end, with real tests and a real API integration — not a static mockup or a Figma prototype.
50	If we deployed this to 500 employees tomorrow, what breaks first?	Concurrency — the single-worker Gunicorn config and synchronous AI calls would need to change before any real-scale rollout.

12. MISSING FEATURES (Prioritized)

Priority	Feature	Est. effort
Critical	Fix the <code>json.stringify</code> → <code>JSON.stringify</code> typo in Prompt Playground	2 minutes
Critical	Stop shipping quiz <code>correct</code> answers to the client before submission	30 minutes
High impact	Upgrade Django 4.2 → 5.2 LTS	1–2 hours (run tests after)
High impact	Add <code>SECURE_SSL_REDIRECT</code> , <code>SESSION_COOKIE_SECURE</code> , <code>CSRF_COOKIE_SECURE</code> , <code>SECURE_HSTS_SECONDS</code>	15 minutes
High impact	Fix rate-limit message/enforcement mismatch	10 minutes
High impact	Add a basic GitHub Actions CI workflow running <code>manage.py test</code>	20 minutes
Medium impact	Escape text before <code>innerHTML</code> in AI Tutor chat (use <code>textContent</code>)	15 minutes
Medium impact	Bump Gunicorn to multiple workers/threads	5 minutes
Medium impact	Add conversation memory (last few turns) to AI Tutor calls	1 hour
Medium impact	Add ARIA labels to icon-only interactive elements	1–2 hours
Nice to have	Tone down or back up the "cryptographically verified" certificate copy with real HMAC signing	1–3 hours
Nice to have	Add README screenshots/GIF + in-repo architecture diagram	30 minutes
Nice to have	Extract duplicated quiz-marker constants into one shared location	20 minutes
Future roadmap	Real RAG over the curriculum for cross-lesson tutor questions	4–8 hours
Future roadmap	A minimal tool-calling demo (curriculum search tool)	3–5 hours
Future roadmap	Organization/Team model + seat management for the B2B pitch	Multi-day, post-hackathon
Future roadmap	Dark mode	2–3 hours

13. FINAL IMPROVEMENT PLAN (4-Day Sprint to Aug 3, 2026)

Priority	Improvement	Reason	Effort	Expected score increase
1	Fix <code>json.stringify</code> typo	Your star demo feature is currently non-functional	2 min	+3-5 pts (avoids a live demo failure entirely)
2	Strip <code>correct</code> answers from client-side quiz JSON	Directly undermines your "verifiable proof of learning" pitch if a judge checks	30 min	+2-3 pts
3	Fix rate-limit message/enforcement mismatch	Small but visible inconsistency a careful judge will catch in testing	10 min	+1 pt
4	Add production security headers (<code>SECURE_SSL_REDIRECT</code> etc.)	Concrete, dated, verifiable via <code>manage.py check --deploy</code>	15 min	+1-2 pts
5	Upgrade to Django 5.2 LTS	Currently running an EOL framework version — a security-auditor judge persona will check this	1-2 hrs	+2-3 pts
6	Add a GitHub Actions CI workflow	Cheap, visible signal of engineering maturity given tests already exist	20 min	+1-2 pts
7	Bump Gunicorn workers/threads	Avoids visible lag if multiple judges test concurrently	5 min	+1 pt (risk mitigation)
8	Escape AI chat message rendering (<code>textContent</code> not <code>innerHTML</code>)	Removes a real, if low-severity, self-XSS vector	15 min	+1 pt
9	Add conversation memory to AI Tutor	Currently can't handle natural follow-up questions — visible in a live demo	1 hr	+2-3 pts
10	Add a minimal tool-calling or RAG demo	Closes the "teaches it but doesn't do it" gap most likely to be probed by strong technical judges	4-8 hrs	+4-6 pts (highest single lever available)
11	Add README screenshots + architecture diagram	Cheap polish, first thing judges see	30 min	+1 pt
12	Add ARIA labels to key interactive icons	Real, fixable accessibility gap	1-2 hrs	+1 pt

Recommended sequencing given only 4 days: Do items 1-4 today (under 1 hour combined, removes all demo-day risk). Do items 5-9 tomorrow. Spend your remaining 2 days on item 10 if you want to seriously compete for top-tier placement — it's the single highest-leverage improvement available and directly addresses the most likely line of tough questioning from a technically sophisticated panel.

14. FINAL VERDICT

Overall score: 71/100 as currently submitted. (Realistically 78-82/100 achievable with the Phase 13 sprint completed.)

Verdict: Shortlist → realistic Finalist. Not currently a Top 10 / winner / grand-prize contender as-is, primarily because of the demo-breaking bug and the "teaches RAG/agents but doesn't build them" gap — both are fixable in your remaining 4 days, and fixing them meaningfully changes this calculus.

10 Biggest Strengths

1. It's a real, fully working, end-to-end deployed application — not a mockup.
2. Real Anthropic API integration with sensible model choice, grounding, and rate limiting.
3. 18 passing automated tests covering genuinely important paths (auth, open-redirect, quiz parsing, certificate issuance).
4. Thoughtful, already-implemented security decisions (server-side quiz re-validation, placeholder API key rejection, tested open-redirect prevention).
5. 60 real, substantial curriculum lessons (11,600+ lines) — real content depth, not filler.
6. Smart caching strategy (lesson-HTML cache keyed on content version).
7. Query-efficient Django ORM usage (`select_related`) (`prefetch_related`) (`annotate`) avoiding N+1 patterns.
8. A genuinely well-produced demo script (`DEMO_PITCH.md`) with a pre-demo seeding command — most teams don't do this.
9. A creative, differentiated packaging (narrow vertical + live prompt sandbox + grounded tutor + public cert verification) even if no single piece is unprecedented.
10. Sensible, environment-driven configuration (secrets via env vars, safe defaults, graceful degradation when the API key is absent).

20 Biggest Weaknesses

1. Prompt Playground is currently non-functional due to a JS typo — your literal star demo feature.
2. Correct quiz/assessment answers leak in page source before submission.
3. "Cryptographically Verified Credential" is marketing language not backed by actual cryptographic signing.
4. Django 4.2 is past end-of-life as of this review.
5. Missing 5 production security settings flagged by Django's own deploy checklist.
6. Rate-limit error messages don't match enforced limits.
7. AI Tutor has no conversation memory across turns.
8. No RAG, agents, or tool calling despite the curriculum teaching all three.
9. No explicit prompt-injection defenses.
10. Self-XSS vector via unescaped `innerHTML` in the AI Tutor chat.
11. No CI/CD despite having a real test suite ready to plug in.
12. Single default Unicorn worker — real concurrency risk during a live multi-judge demo.
13. No conversation/response quality evaluation framework for the AI features.
14. No accessibility attributes (ARIA) anywhere across 15 templates.
15. No dark mode.
16. Enterprise B2B features (teams, seats, SSO) are pitched but not implemented.
17. Thin competitive moat — easy to clone the core mechanics.
18. `.env.example` ships with `DEBUG=True`.
19. No Subresource Integrity on third-party CDN scripts (Tailwind, PrismJS, Mermaid).
20. Duplicated quiz-marker-detection logic across three locations in `views.py`.

20 Highest-Impact Improvements

1. Fix the `json.stringify` typo.

2. Strip client-visible quiz answer keys.
3. Upgrade Django to 5.2 LTS.
4. Add missing production security headers.
5. Fix rate-limit message/enforcement mismatch.
6. Add a GitHub Actions CI workflow.
7. Bump Gunicorn worker/thread count.
8. Escape AI chat rendering (`{textContent}`).
9. Add short-term conversation memory to the AI Tutor.
10. Build even a minimal RAG or tool-calling demo to close the "teach it, don't do it" gap.
11. Either implement real HMAC certificate signing or soften the "cryptographic" language.
12. Add README screenshots and an in-repo architecture diagram.
13. Add ARIA labels to icon-only controls.
14. Add SRI hashes to CDN script tags.
15. Consolidate duplicated quiz-marker constants.
16. Add a lightweight prompt-injection guardrail note/test to the Tutor system prompt.
17. Fix `{env.example}` to ship `{DEBUG=False}`.
18. Add composite DB indexes on `{UserProgress}/{QuizAttempt}` for future scale.
19. Add basic AI response logging for future quality evaluation.
20. Scope and stub even a minimal "Team" model to make the B2B pitch demonstrable, not just narrated.

Would I, personally, vote for this project?

Yes — conditional on fixing items 1–4 in the Phase 13 sprint before demo day. As-is, with a broken headline feature, I could not in good conscience rank it above projects that demo cleanly, no matter how solid the code underneath. But the underlying engineering — real tests, real security instincts already partially applied, a genuinely working AI integration, and a legitimately large body of real content — is stronger than the median hackathon submission I'd expect to see. Fix the four cheap items today, and this becomes an easy, confident vote for shortlist-to-finalist placement, with a real shot at higher if you also land the RAG/tool-calling upgrade.

15. RISK MATRIX

Risk	Likelihood	Impact	Mitigation
Prompt Playground fails live in front of judges	Certain (100%) until fixed	Severe — it's your handed-to-a-judge centerpiece	Fix the 1-line typo today; re-test before demo day
Judge inspects page source and finds quiz answers	Moderate	Moderate — undermines your certification-integrity pitch	Strip <code>correct</code> field from client JSON
Security-minded judge runs <code>manage.py check --deploy</code> or asks about Django version	Moderate-High given your stated judge personas include a "security auditor"	Moderate	Complete items 3-5 in Phase 13
Technical judge asks why a course teaching RAG/agents doesn't use them	High given stated Google/OpenAI-caliber judge personas	Moderate-High on scoring	Ship even a minimal RAG/tool demo if time allows
Concurrent judges cause visible lag on AI endpoints	Moderate	Low-Moderate	Bump Unicorn workers before demo
Judge opens <code>.meta/</code> folder and questions authorship/ownership	Low-Moderate	Low if you have a confident answer ready	Prepare the on-theme answer in §9, or remove the folder

16. WINNING PROBABILITY

- **As submitted right now (with the Playground bug live):** Low-to-moderate. A broken headline demo feature in front of judges is a hard, visible failure that overshadows genuinely good underlying work.
- **After the ~1 hour of Priority-1 fixes (items 1-4, Phase 13):** Solidly competitive for shortlist/finalist tiers in a general-purpose hackathon track.
- **After the full 4-day sprint including the RAG/tool-calling addition:** Realistic contention for top-tier placement, though "grand prize" against genuinely novel multi-agent or frontier-research submissions from top-tier events remains a stretch given the current AI architecture's simplicity.

17. ACTION CHECKLIST (do these in order)

- ☐ Fix `json.stringify` → `JSON.stringify` in `templates/courses/prompt_playground.html` line 116
- ☐ Remove/strip the `correct` field from `daily_quiz_json` and `questions_json` before they reach the template
- ☐ Fix rate-limit error message text to match the actual 100/hour enforcement (or lower the enforced number to match the message — pick one)
- ☐ Add `SECURE_SSL_REDIRECT`, `SESSION_COOKIE_SECURE`, `CSRF_COOKIE_SECURE`, `SECURE_HSTS_SECONDS` to `settings.py`
- ☐ Upgrade `Django==4.2.11` → `Django>=5.2,<5.3` in `requirements.txt`, re-run all 18 tests
- ☐ Add `.github/workflows/tests.yml` running `python manage.py test`
- ☐ Update `Procfile` to `gunicorn ai_course_platform.wsgi --workers 3 --threads 2 --log-file -`

- ☐ Replace `innerHTML` with `textContent` for chat message rendering in `lesson_detail.html`
- ☐ Add last-N-turns conversation history to the AI Tutor's `messages` array
- ☐ Decide: soften "cryptographically verified" copy, or implement real HMAC signing
- ☐ Add 1–2 screenshots and a GIF to `README.md`
- ☐ Rehearse the full 8-minute demo script end-to-end at least twice before Aug 3